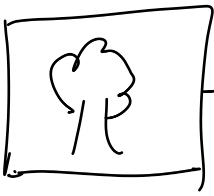
Basics of Information Theory



→ Classification network → $\begin{cases} \text{Tree } 90\% \\ \text{Cat } 5\% \\ \text{dog } 5\% \end{cases}$ ⇒ Tree

(Training a probability model)

⇒ Target: Minimizing the difference between the resulting distribution and "real distribution" ⇒ minimizing cross-entropy Loss

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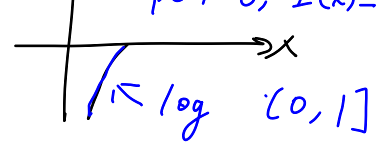
1. Self Information: Consider a random variable X, with target x, probability distribution $p(x)$. The self information when $X=x$ is defined as:

$$I(x) := \underline{-\log p(x)}$$

$p(x)\uparrow$ self information $\downarrow$

$p(x)=1,\ I(x)=0$

$p(x)=0,\ I(x)=+\infty$



unit of $I(x)$: $\begin{cases} 2 \text{ as the base of logarithm; bit} \\ e \text{ ....: nat} \end{cases}$

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2. Entropy: Expectation of Self information.

discrete

$$\underline{H(x)} = E_X[I(x)] = E_x[-\log(p(x))] = \underline{-\sum_{x\in\mathcal{X}} p(x)\log(p(x))}$$

note: in case $p(x_i)=0$, $0\log 0 := 0$ $\quad \lim_{p\to 0^+} p\log p = 0$

$H(x)\uparrow$, more information (possibility) is contained,

⇒ ① deterministic information: $H(x)=0$

② uniform distribution: $H(x)$ is maximal.

Example

$P(X_1)$	$P(X_2)$	$P(X_3)$	Entropy
1	0	0	0
1/2	1/4	1/4	$\frac{3}{2}\log 2$
1/3	1/3	1/3	$\log 3$ (max)

3. Joint Entropy and conditional Entropy

Consider random variables X and Y with target set $\mathcal{X}$ and $\mathcal{Y}$, respectively.

① Joint Entropy: $H(\underline{X},\underline{Y}) := -\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(x,y)\log(p(x,y))$

② conditional Entropy: $H(X|Y) := -\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(x,y)\cdot\log(p(x|y))$ $\quad p(x,y)=p(x|y)\cdot p(y)$

$$= -\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(x,y)\cdot\log\frac{p(x,y)}{p(y)}$$

$$= \underbrace{-\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(x,y)\log(p(x,y))}_{H(X,Y)} - \Big(\underbrace{-\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(x,y)\cdot\log(p(y))}\Big)$$

$\sum_{x\in\mathcal{X}} p(x,y) = p(y)$

$-\sum_{y\in\mathcal{Y}} p(y)\log(p(y)) \longrightarrow H(Y)$

$$H(X|Y) = \boxed{H(X,Y) - H(Y)}$$

③ Mutual Information: the uncertainty of one random variable given another random variable is fixed.

$p(x,y)=p(x)\cdot p(y)$

$\log = 0$

$(X, Y$ are independent$)$

$$I(X;Y) = \sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(x,y)\cdot\log\frac{p(x,y)}{p(x)\cdot p(y)}$$

Another angle: $I(X;Y) = H(X) - H(X|Y)$

$$= H(Y) - H(Y|X)$$

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4. Cross Entropy: the amount of information when using $q(x)$ to approximate $p(x)$ ← (real distribution)

used as approximation)

$$H(p,q) = E_p[-\log(q(x))] = -\sum_x p(x)\log(q(x))$$

← information contained in $q(x)$.

q and p is closer, $H(p,q)\downarrow$, p and q is further, $H(p,q)\uparrow$

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5. Kullback-Leiber Divergence (KL-Divergence.

$$KL(p,q) := H(p,q) - \underline{H(p)}$$

$$= -\sum_x p(x)\log(q(x)) + \sum_x p(x)\log(p(x)) = \sum_x p(x)\log\frac{p(x)}{q(x)}$$

Statement: $KL(p,q)\geq 0$ and $KL(q,p)=0$ if $p(x)=q(x)$.

Proof: For $x\in(0,1]$, one has $\log(x)\leq x-1$, $=$ holds iff $x=1$

$$-KL(p,q) = \sum_x p(x)\log\frac{q(x)}{p(x)} \leq \sum_x p(x)\Big(\frac{q(x)}{p(x)}-1\Big) = \sum_x (q(x)-p(x))$$

$= \underbrace{\sum_x q(x)}_{1} - \underbrace{\sum_x p(x)}_{1} = 0 \implies -KL(p,q) \le 0 \implies KL(p,q) \ge 0.$

Note: ① $KL(p,q) = \sum_x p(x) \log \frac{p(x)}{q(x)} = \sum_x p(x) \log(p(x)) - \sum_x p(x) \log(q(x)) \ge 0$

$\implies \sum_x p(x) \log(p(x)) \ge \sum_x p(x) \log(q(x))$

$\implies -\sum_x p(x) \log q(x) \ge -\sum_x p(x) \log(p(x))$ (Gibbis inequality).

② KL-divergence (relative entropy) characterize the information loss using distribution q to approximate p.

③ Similar to cross divergence, q and p is closer, $KL(p,q) \downarrow$ (characterize "distance" between q and p)

↳ NOT a real distance $\begin{cases} \text{symmetric} \\ \text{triangular inequality} \end{cases}$

Exercise: a discrete distribution with three event X_1, X_2, X_3

real distribution: P:

$$P(X_1) = \frac{1}{2} \quad P(X_2) = \frac{3}{10} \quad P(X_3) = \frac{1}{5}.$$

A model predict a distribution Q

$$Q(X_1) = \frac{1}{5} \quad Q(X_2) = \frac{2}{5} \quad Q(X_3) = \frac{2}{5}.$$

KL-divergence $KL(P,Q) =$ ____________ (nat),

$KL(P,Q) = H(P,Q) - H(P)$

$H(P,Q) = -\sum_x p(x) \ln(q(x)) = -\left(\frac{1}{2}\ln\frac{1}{5} + \frac{3}{10}\ln\frac{2}{5} + \frac{1}{5}\ln\frac{2}{5}\right)$

$H(P) = -\sum_x p(x) \ln(p(x)) = -\left(\frac{1}{2}\ln\frac{1}{2} + \frac{3}{10}\ln\frac{3}{10} + \frac{1}{5}\ln\frac{1}{5}\right)$

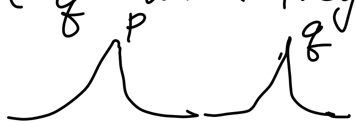
6. Jensen-Shannon Divergence

$$JS(p,q) := \frac{1}{2}KL(p,m) + \frac{1}{2}KL(q,m), \text{ with } m = \frac{1}{2}(p+q)$$

(a symmetric quantity characterizing similarity between two distribution) → NOT satisfying triangular inequality.

Remark: Common drawback of KL and JS divergence

Difficult to to characterize p and q when they have little or no intersection.



① KL Divergence: $\boxed{p(x) > 0, \quad q(x) = 0}$ $\quad \log \frac{p(x)}{q(x)} \to +\infty$

$$KL(p, q) \to +\infty$$

② JS Divergence:

$$\nearrow \frac{p(x)}{p(x)/2}$$

$$JS(p,q) = \frac{1}{2} \sum_x p(x) \log \frac{p(x)}{(p(x)+\underbrace{q(x)}_{=0})/2} + \frac{1}{2} \sum q(x) \log \frac{q(x)}{(\underbrace{p(x)}_{=0}+q(x))/2}$$

$$= \frac{1}{2} \underbrace{\sum_x p(x)}_{1} \log 2 + \frac{1}{2} \underbrace{\sum_x q(x)}_{1} \log 2$$

$$= \log 2.$$

7. Wasserstein Distance

Consider distribution q_1, q_2, the p-th Wasserstein distance is

$$W_p(q_1, q_2) = \inf_{\gamma(x,y) \in \Gamma(q_1, q_2)} E_\gamma [d(x,y)^p]^{\frac{1}{p}}$$

$\Rightarrow$ Fulfilling the def of distance thus more robust during the training process